Building a Wildfire Smoke Dataset for Early Smoke Detection

Data Curation, Annotation, and Preparation for a Deep Learning Model

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***Abstract*— In this project, we created a curated and labeled image dataset from about 1,000 wildfire surveillance videos for potential use in training machine learning models to detect wildfire smoke. The dataset includes both positive and negative smoke instances, with smoke images further categorized by density (low, medium, high). The process involved cleaning raw video data, extracting high-quality frames, manually labeling smoke presence and density, and formatting the data for compatibility with TensorFlow API. The result is a structured, accessible dataset suitable for future computer vision applications in wildfire detection.**

Keywords— Dataset, Machine Learning, Wildfire Detection, TensorFlow API, Computer Vision

I. Introduction

Throughout the world, climates are becoming increasingly arid. Likely a result of global warming and rising temperatures, the aftereffects are being seen throughout the country in the form of wildfires. Over the years, the number of wildfires has been steadily increasing, leaving land and homes to be burned down as shown in Fig. 1 and Fig. 2. As wildfire events continue to grow in frequency and severity, early detection has become a key component of disaster response and prevention. Many existing detection systems rely on satellite data or manual monitoring, which can delay response times. This project aims to contribute to more efficient detection systems by preparing a labeled dataset suitable for training machine learning models, especially convolutional neural networks, to recognize smoke in real-time surveillance footage. Our goal was not to train a model, but to build a foundational resource others can use in model development.

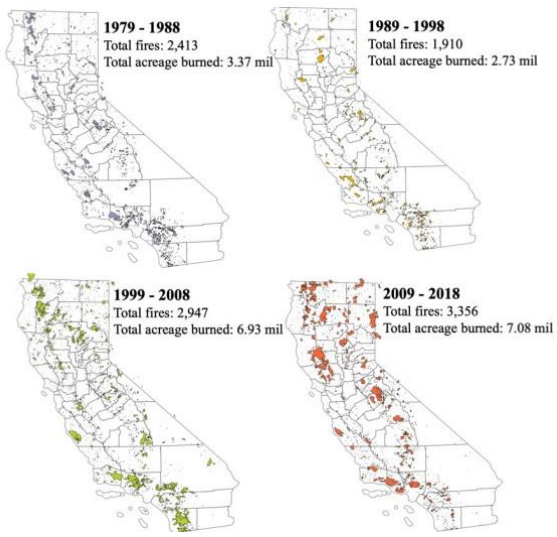

Fig. 1: Mapping of wildfires in California [1].

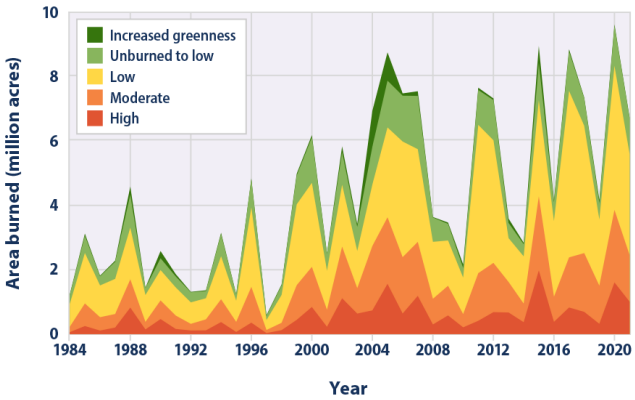

Fig. 2: Number of acres burned in America over the years [2].

II. Background and Motivation

Wildfires cause devastating damage across ecosystems and communities, often resulting in loss of life, destruction of property, and long-term environmental degradation. As climate change continues to intensify, the frequency, scale,

and unpredictability of wildfires have increased significantly. Detecting these events at an early stage has become critical for timely evacuation, resource deployment, and reducing destruction.

Traditional wildfire detection systems such as manual lookout towers, ground patrols, and satellite imaging, often suffer from limited coverage and delayed response time. These methods can miss small or early-stage smoke plumes, especially in remote or forested areas resulting in action not being taken until it is too late. In contrast, computer vision systems powered by machine learning, particularly convolutional neural networks (CNNs), offer the potential for real-time smoke detection through automated analysis of video streams.

However, the performance of such models is heavily dependent on the quality of the datasets used during training. Unfortunately, there is a shortage of publicly available, large-scale, and well-annotated wildfire smoke datasets. Many existing datasets are for detecting the fire itself which defeats the purpose of catching fires before they become too destructive. And any actual smoke datasets are limited in size or captured in ideal conditions which fails to take into acount the condition of many actual wilderness cameras. This limits model generalizability and performance when deployed in practical settings.

This project is motivated by the need to bridge this gap by constructing a high-resolution, labeled dataset derived from real, time-lapse wildfire surveillance videos. These videos capture diverse environmental conditions. This includes varying smoke densities, weather patterns, lighting conditions, and obstructions which are critical for developing robust detection models. By focusing on manual annotation and careful data filtering, we aim to create a reliable, structured dataset that serves as a foundation for future research in automated wildfire smoke detection.

III. Approach

The objective of this project is to prepare a dataset of wildfire smoke videos for use with TensorFlow or PyTorch by extracting relevant frames and labeling them as either containing wildfire smoke or not wildfire smoke, while also categorizing the density of the smoke. The process can be divided into several key phases: data collection and cleaning, video labeling and frame extraction, dataset organization, and dataset creation for TensorFlow compatibility. Through these steps, we will have a dataset complete with wildfire smoke images of different densities and environments.

A. Data Collection and Cleaning

The raw data provided for this project consisted of 962 video files, each approximately one minute long. It is important to note that the videos were fast-forwarded in time, meaning that each second in the video represented a much longer period in real time. This posed an initial challenge in terms of interpreting the footage correctly and ensuring that the extracted frames were of high quality.

To begin, the video files were downloaded from the online repository provided (via Dropbox). The raw video files were collected from various Nevada wilderness cameras both around northern Nevada and Lake Tahoe. This provided some diverse scenery and environments as many videos showcased the vast Nevada deserts and mountains while others introduced far more greenery into the dataset through Tahoe's forests. Given the large number of videos, the first task was to assess the overall quality of the data. Many of the video files contained footage that was blurry, unstable, or unclear, likely due to poor camera quality or unstable environmental conditions. These problematic values were handled at the same time as the labeling and frame extracting through a process discussed in the next step.

B. Labeling and Frame Extraction

Now that the raw data is collected, the next step involves removing any low-quality data, while also labeling frames extracted throughout each video. To facilitate this process, a Python-based graphical user interface (GUI) was developed. This GUI provided a user-friendly interface for reviewing the videos and automatically extracting frames at regular intervals. Each frame was visually inspected, and the user assigned labels based on the presence of wildfire smoke and its density. There was also a feature that allowed the specific frame to be skipped altogether in the case of bad quality, lighting, or obstructions. This feature allowed the use of videos that were overall low quality but still provided some high-quality frames. The GUI allowed for rapid navigation through the frames, enabling efficient labeling and extraction. The density classification was performed visually, with the smoke density being categorized based on subjective judgment.

In total, the GUI allowed the user to label a large number of frames quickly and consistently, significantly speeding up the labeling process. For each labeled frame, the image was saved as a separate file, and its associated label (wildfire smoke or not, along with density classification) was recorded in a CSV file. One such extracted image can be seen in Fig. 3. The naming convention for the images included both the video number as it was loaded into the GUI and the frame number for that video. This step was crucial to the overall dataset creation process and was made significantly faster through the GUI created as it allowed supervised labeling while adding some autonomy into the process. The labeling process was still time-consuming but necessary to create a high-quality dataset, as the accuracy of the final model would depend heavily on the quality and consistency of the labels.

Fig. 3: Extracted frame from video using GUI.

C. Dataset Organization

While each frame was being labeled, it was also being saved into a frame directory with its name being changed to the video and frame saved by the GUI labeler. The labels themselves were stored in a reciprocal CSV file using Excel that stored the name of the image being saved to file and whether the image had smoke and its perceived smoke density. The importance of this stage comes from consistently naming the saved images and their labels. This allows a simpler data pipeline for converting the images into a TensorFlow dataset. Another thing considered in this step was the possibility of under sampling or over sampling due to class imbalances. This was an especially prevalent problem with this data since all the videos in the provided Dropbox were of wildfires, which in turn would result in our dataset having far more images of wildfires than not.

D. Dataset Creation for TensorFlow

With the labeled images organized and ready for use, the next step was to convert the images into a format suitable for TensorFlow. TensorFlow provides an API for creating custom datasets, which can then be used directly for model training. This dataset creation process involved loading the labeled images into TensorFlow's tf.data.Dataset format, which allows automatic preprocessing that turns the data into a TensorFlow dataset.

Using TensorFlow's tfds.core.GeneratorBasedBuilder and other related functions, the labeled images were automatically loaded and converted into a structured dataset. These functions read the images from their directory and match the correct CSV label to each image in order to create the dataset. A key part of this process involved resizing and normalizing the images to ensure that they could be processed efficiently by a model.

The dataset was then split into three main subsets: a training dataset, evaluation dataset, and testing dataset. The training dataset contained the majority of the labeled data and was used to train the model, while the evaluation dataset consisted of a smaller subset used for validation and performance assessment. Finally, the testing dataset remains completely unseen by the model in order to determine accuracy and prediction rates. To create these subsets, a 70-15-15 split was applied, with 70% of the data reserved for training, 15% for evaluation and the last 15% for testing. The splitting was done randomly to ensure that the evaluation set was representative of the full dataset and that the model could generalize well to unseen data.

The final dataset, now in the form of a TensorFlow-compatible dataset object, was ready for use in model training and evaluation. This dataset was structured to provide efficient access to data during model training, evaluation and testing, allowing the machine learning pipeline to be streamlined for faster model experimentation.

IV. Evaluation

Following the data labeling and dataset creation pipeline, a comprehensive set of statistics was gathered to assess the effectiveness and quality of the resulting wildfire smoke image dataset. The original dataset consisted of 962 high-definition (1920×1080) video files, each approximately one minute in length. From each video, four evenly spaced frames were extracted to ensure temporal coverage and reduce redundancy, yielding a total of 4,280 raw frames.

Utilizing the created GUI, frames with poor visibility, significant camera overlays, or instability (e.g., blur, extreme lighting, or corrupted footage) were removed from the dataset. After this filtering stage, 1,938 high-quality frames were retained for labeling and dataset construction. The labeled dataset includes the following class distributions.

Smoke labels:

- Smoke present: 1,290 images
- No smoke: 648 images

Smoke density breakdown:

- Low: 477 images
- Medium: 492 images
- High: 321 images

These labeled categories help capture not only the presence of smoke but also the severity of its visual impact, which is critical for training robust detection models.

Several challenges were encountered during the labeling process. Footage captured at night significantly reduced the visibility of smoke, making it difficult to accurately classify such frames. Atmospheric conditions such as rain, fog, or lens flare often mimicked the visual characteristics of smoke, leading to ambiguity in some cases. In addition, the presence of tall structures, poles, or other camera obstructions frequently obscured parts of the frame, complicating manual judgment. Distant smoke plumes were also sometimes difficult to distinguish from clouds or haze especially in poor weather conditions.

Despite these limitations, the resulting dataset provides a diverse and well-annotated collection of wildfire-related imagery as can be seen in Fig. 4. It represents a range of environmental conditions, smoke densities, and visibility challenges, making it well-suited for training and evaluating deep learning models using TensorFlow or PyTorch frameworks. The dataset was also split into training, validation, and testing subsets to support future wildfire detection research.

Fig. 4: Preprocessed and labeled images from the training split.

V. Discussion

While our dataset lays a solid foundation for training machine learning models in wildfire smoke detection, several challenges emerged throughout the process.

Determining smoke presence and density was inherently subjective. Factors like lighting conditions, camera angles, and environmental elements (e.g., fog, dust) often made it difficult to distinguish smoke accurately. This subjectivity could introduce inconsistencies in the dataset, potentially affecting model performance.

Given that the source videos predominantly featured wildfire events, there was a natural skew towards frames containing smoke. This imbalance poses a risk of models becoming biased, potentially leading to higher false-positive rates when deployed in real-world scenarios where smoke is less prevalent.

The dataset encompasses a range of terrains and weather conditions, from Nevada's arid landscapes to Tahoe's lush forests. While this diversity is beneficial, it also introduces variability that models must learn to generalize across. Additionally, nighttime footage presented challenges due to reduced visibility, complicating smoke detection.

Our dataset aligns with efforts like Pyro-Near 2024, which emphasizes real-world smoke detection challenges. However, unlike some datasets that incorporate synthetic images to augment data, our approach relied solely on real surveillance footage, aiming for authenticity but potentially limiting variability.

Despite these challenges, the dataset's structured format and detailed annotations make it a valuable resource for developing and testing machine learning models. It offers opportunities to explore techniques like data augmentation, transfer learning, and domain adaptation to enhance model robustness.

VI. Related Works

The field of wildfire detection has seen significant advancements with the integration of machine learning and computer vision techniques. One notable contribution is the PyroNear-2024 dataset, which offers a comprehensive collection of annotated images and videos of smoke plumes captured in real-world conditions. This dataset encompasses approximately 50,000 images with 150,000 manual annotations, covering 400 wildfire events across regions like France, Spain, and the United States. Its diversity and scale make it a valuable resource for training and evaluating smoke plume detection models, including those that process sequential data. The dataset's challenging nature, as evidenced by cross-dataset experiments yielding an F1 score of around 60%, underscores the complexity of real-world wildfire detection tasks.

In addressing the issue of data imbalance in wildfire image datasets, researchers have explored data augmentation techniques. A study published in MDPI's Remote Sensing journal introduced a method combining DenseNet, a type of convolutional neural network, with CycleGAN-based data augmentation. This approach generated synthetic wildfire images to balance the dataset, leading to improved model performance. The integration of CycleGAN allowed for the creation of realistic fire images, enhancing the training process and resulting in higher accuracy and F1 scores in wildfire detection tasks.

These works highlight the importance of diverse and well-annotated datasets, as well as innovative data augmentation methods, in advancing the capabilities of machine learning models for early wildfire detection.

VII. Conclusion and Future work

In this project, we successfully curated a labeled dataset from approximately 1,000 wildfire surveillance videos, extracting and annotating frames to indicate the presence and density of smoke. This dataset serves as a foundational tool for training machine learning models aimed at early wildfire detection.

There is still much room for improvement, and the question of what step to take next. To enhance the quality of the dataset in the future, we would suggest implementing semi-supervised or unsupervised learning techniques, as they could assist in labeling vast amounts of data more efficiently, reducing manual effort and potential subjectivity.

Additionally, incorporating synthetic data generation methods, such as Generative Adversarial Networks (GANs), can help address class imbalances and introduce variability, enhancing model training. We could also focus on model accuracy; combining visual data with other modalities, like thermal imaging or sensor data, could improve detection accuracy, especially under challenging conditions like low visibility or nighttime scenarios, which are not uncommon in a real-life setting.

After these improvements are made, collaborating with firefighting agencies to test models in operational settings would provide insights into practical challenges and model performance in dynamic environments. This would then allow us to test how it functions in the field and make even more significant improvements to not only the model itself, but also how we are utilizing it.

By addressing these areas, we aim to refine our dataset further and contribute to the development of more effective wildfire detection systems, aiding in timely responses to wildfire outbreaks.

References

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